

WASP-11b

R-0700

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_241230 · created 2026-07-27T21:43:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460674.7135 +/- 0.0026 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.011
Transit depth (Rp/Rs)^2	2.41 +/- 0.35 [%]
Orbital Inclination (inc)	87.44 +/- 0.8
Airmass coefficient 1 (a1)	1.159 +/- 0.0084
Airmass coefficient 2 (a2)	-0.122 +/- 0.022
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	None
Optimal Aperture	4.64
Optimal Annulus	5.78
Transit Duration (day)	0.0745 +/- 0.0058

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.011 vs 0.131 (deviation 1.65 σ)
Duration fit vs published	0.0745 d vs 0.1075 d (deviation 4.29 σ)
Mid-time O-C	9.7 min
Depth SNR	10.6
Residual scatter	0.72 %

Light curve

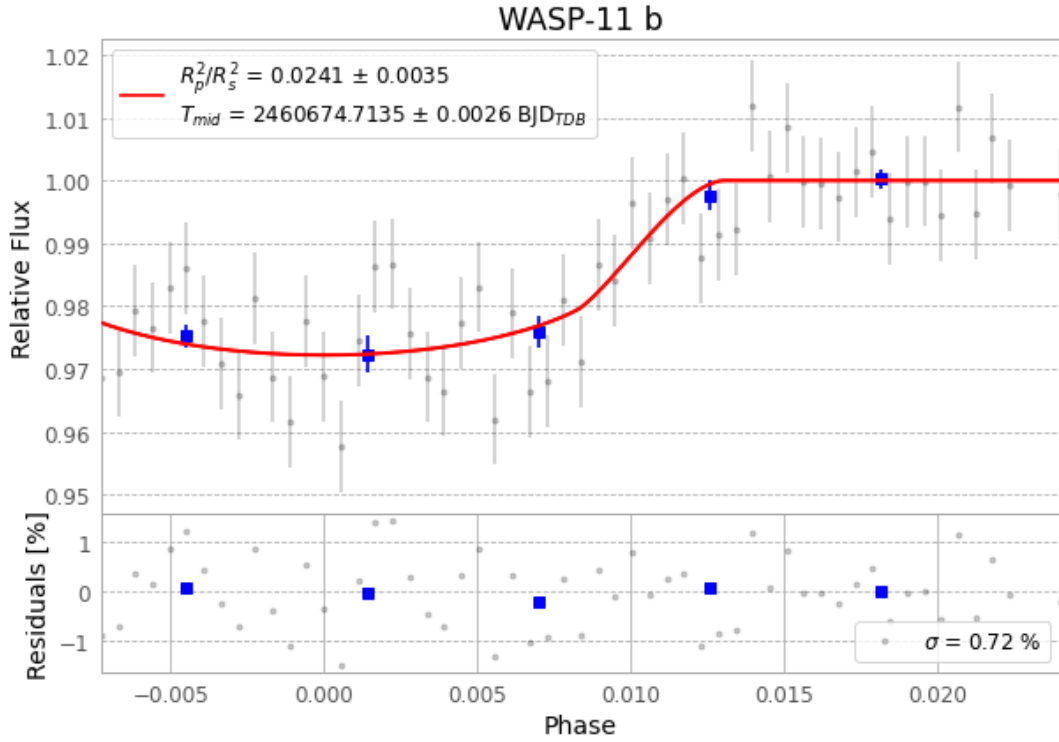


Figure 1 — FinalLightCurve_WASP-11 b_2024-12-29.png (SHA-256 7fdac10d86e855d7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [381, 259], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9f83ce6c289e928...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 62f7813

Data provenance

55 FITS frames (36 MB), HATP-10241230042116.FITS → HATP-10241230070912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-241230.log (c6c0d232a0f5...), FinalLightCurve_WASP-11 b_2024-12-29.png (7fdac10d86e8...), FinalLightCurve_WASP-11 b_2024-12-29.csv (798d97d3c657...)

Compute resources

Wall time 95.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-27 21:43Z.